

Every Constructor an Interaction

Observer extensions, and the discriminating power
of spatial formulae in interactive graph-structured lambda theories

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Abstract

Spatial logics separate more than bisimulation does. This is not a defect and it is not news: Caires observed two decades ago that the equivalence induced by a spatial logic for the π -calculus sits strictly between structural congruence and strong bisimilarity, and the intensionality of such logics was settled shortly afterwards by Sangiorgi and by Hirschhoff, Lozes and Sangiorgi. Yet the same observation, restated, is routinely taken to refute any claim that a generated spatial-behavioural logic is *adequate* — that its formulae separate exactly what interaction separates. The two claims look incompatible and are not.

They are not incompatible because there is no single observational bisimulation against which a spatial logic must be measured. Observational equivalence is indexed by the destructuring capabilities admitted to the observer. We make the index explicit and mechanical. Given an interactive graph-structured lambda theory $\mathbb{G}$ with interaction constructor K , and a set $\mathcal{O}$ of constructors the observer may take apart, we adjoin one nullary probe atom per constructor and per argument position, freely adjoin a correctly sorted argument former per signature profile, and overload K so that a probe meets a term. Three rewrite rules per constructor then make asking, projecting and rebuilding into interactions. In the resulting theory $\text{Obs}(\mathbb{G}, \mathcal{O})$ the observer's structural questions are ordinary transitions, and bisimilarity descends monotonically from context-labelled bisimilarity on $\mathbb{G}$ at $\mathcal{O} = \emptyset$ to the equational theory of $\mathbb{G}$ at $\mathcal{O} = \Sigma$, which is where the spatial connectives already were. The two disputed propositions are the two endpoints of one family.

The phenomenon is folklore in the spatial logics community, where extensionality of spatial observation has been established *semantically*, by exhibiting models — typically distributed calculi with failures — whose observational equivalence is characterised by a spatial logic. Our handling is *syntactic*: a transformation of presentations, mechanical in the signature. We prove a reconstruction theorem for the operational side and a separation theorem for the logical side, keeping them apart, so that a weakness in one does not look like a weakness in both; and we record what the construction costs. It is not object-capability safe, since unrestricted opening destroys encapsulation. We describe the shape of the capability discipline that repairs this, and do not develop it here.

The note is downstream of [1], in preparation, which supplies the label discipline: labels are minimal contexts enabling a rewrite, in the sense of Leifer and Milner, in an open form that does not restrict to ground terms.

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1 Introduction

1.1 The phenomenon

A spatial-behavioural logic for a process calculus carries two layers. The behavioural layer is modal: formulae speak of the steps a term can take. The structural, or spatial, layer speaks of how the term is put together — that it is a parallel composition of two parts with such-and-such properties, that it is a send on a name, that it is empty.

The layers do not have the same separating power, and the direction of the gap surprises people the first time they meet it. Behavioural equivalence is coarse by design: it identifies terms that no experiment can tell apart. Structural predicates are finer, because they can ask questions no experiment can pose — or rather, no experiment *in the calculus as presented*. That qualification is the subject of this note.

1.2 The apparent contradiction

When the logic is not designed but *generated* from the presentation of the calculus — one structural connective per term former, one modality per rewrite rule and redex position — one wants an adequacy theorem: logical equivalence coincides with the calculus’s observational equivalence. Adequacy is what makes a generated logic worth generating. Too coarse and there are observable distinctions the logic cannot state; too fine and the logic invites its user to chase differences no experiment could settle.

So one wants to assert both of:

$$\text{logical equivalence} = \text{observational equivalence}, \quad (\text{A})$$

$$\text{the spatial layer} \subsetneq \text{observational equivalence}. \quad (\text{B})$$

Read carelessly these are inconsistent, and a careful reader will say so. We have seen (B) offered as a refutation of (A) more than once, by readers who knew the literature well enough to recognise (B) and not well enough to know what answers it.

1.3 The resolution

There is no such thing as *the* observational equivalence of a calculus independent of what the observer is permitted to do. (B) is a statement about bisimilarity in $\mathbb{G}$, where the observer may build contexts out of $\mathbb{G}$ ’s constructors and watch what happens. (A) is a statement about bisimilarity in a theory in which the observer additionally holds instruments for taking terms apart. Exhibit the index and the family interpolates between them; there is then nothing to reconcile.

The construction is the subject of §4. Where the cut has a transparent unit its slogan is:

Make every term former an interaction. Bisimulation observes interactions. Therefore bisimulation observes every term former.

1.4 What is new here, and what is not

The phenomenon is known. §2 sets out four lineages of prior work, of which one — extensionality of spatial observation — already establishes that spatial logics need not be counted intensional, and another — reactive systems — supplies the label discipline we use. We claim no priority over any of it, and the underlying idea of reifying constructors as tags with observation actions is elementary enough that we expect precedents we have not found.

What we add is a change of method, and the change buys generality. The prior extensionality results are *semantic*: they exhibit a calculus, typically a distributed one with local and remote communication and partial failures, and show that its observational equivalence is characterised by a spatial logic. The extra separating power comes from features of that calculus, which are modelling choices about a particular domain.

Our construction is *syntactic*. It is a transformation of presentations. It reads the signature, adjoins probe atoms and argument formers, adds three rules per constructor, and stops. It is defined for any interactive GSLT satisfying the hypotheses of §3 rather than for one calculus, and because it is mechanical in the presentation it composes with the machinery that generates the logic from the same presentation. That last point is what matters for our purposes: if both the logic and the enlargement are functions of the signature and of $\mathcal{O}$, then adequacy is the statement that two functions of the same argument agree, rather than a coincidence to be checked calculus by calculus.

1.5 Three things kept apart

An earlier draft of this note braided three claims together tightly enough that a weakness in one looked like a weakness in all three. We separate them here and recommend the separation to anyone reasoning in this area:

- **operational reconstruction** — what the enlarged transition system can distinguish (§5);
- **logical expressiveness** — what the generated structural language can distinguish (§6);
- **observer capability** — which instruments a given observer actually holds (§8, §10).

1.6 What this note does not claim

It does not claim that structural inspection was interaction all along. That would be slippery, and an earlier draft said it. The form of observation remains transition observation, but the set of observable transitions has been enlarged, and the enlargement is exactly what supplies the additional discriminating power. The honest statement is that structural observation *can be represented as* interaction, after conservatively extending the observer’s transition theory with destructuring actions.

It does not develop the capability discipline of §10. It does not treat weak equivalences or quantitative refinements. And it makes no claim that unrestricted opening is a sensible thing to permit in a deployed system.

2 Four lineages

Much of the confusion we are addressing comes from reading distinct strands of prior work as one.

The observation. Caires, studying an expressive spatial logic for the synchronous π -calculus with recursion, gives coinductive and equational characterisations of the equivalence the logic induces and locates it: strictly between structural congruence and strong bisimilarity [7]. The companion development with Cardelli [8, 9] is explicit that formulae are taken to denote processes up to structural equivalence rather than up to bisimulation, precisely because a tensor operator that distinguishes bisimilar processes is difficult to reconcile with the latter. This is (B), stated by the people who first had to confront it.

strand	question settled	what it does not settle
the observation	where the induced equivalence sits relative to $\sim$ and $=_E$	whether the gap is intrinsic
intensionality	that the induced equivalence is essentially $=_E$; which connectives are eliminable	whether an observer could be given the missing power
extensionality	that in suitable models observational equivalence is characterised spatially	whether this is obtainable uniformly from a presentation
reactive systems	how to derive labels making bisimilarity a congruence	what happens when the observer is given destructors

Table 1: The four strands, and the question each leaves open. This note addresses the right-hand cells of the last two rows.

The intensionality results. Sangiorgi [10] and then Hirschhoff, Lozes and Sangiorgi [11, 12] settle where the induced equivalence sits: for logics of this kind it is essentially structural congruence. Lozes adds the elimination results — that in the static ambient logic the adjuncts contribute no expressive power [13], later strengthened by Dawar, Gardner and Ghelli through model-comparison games [16] — and the picture is completed by Caires and Lozes [15], who show that a minimal dynamic spatial logic already encodes action quantification and action modalities, so that the spatial and modal fragments have the same separating power, coinciding with structural congruence modulo permutation of actions. Lozes’ thesis [14] is the systematic account. This strand says the logic is *intensional*: it sees syntax. It is often read, wrongly, as saying that nothing can be done about this.

The extensionality results. Something can be done about it. Caires and Vieira [17] address the tension directly, showing that there are natural models in which extensional observational equivalences *are* characterised by spatial logics, composition and void included — and concluding that spatial observations need not always be counted intensional, even when expressive enough to describe structure. Their development is carried out in a minimalist distributed calculus with local synchronous communication, local computation, asynchronous remote communication, and partial failures. The additional separating power available to an observer in that setting is what closes the gap. A related model-theoretic treatment is given by Tuosto and Vieira [18]; Caires’ tutorial survey [19] is the best single entry point.

Reactive systems. Labelled transition systems were for a long time built by hand, the researcher choosing labels that would give the right view of behaviour. Leifer and Milner [2] showed how to derive them: labels should be the *minimal contexts* enabling a rewrite, minimality determined by a universal property, and bisimilarity over the derived transition system is then a congruence by construction rather than by contextual closure after the fact. Sewell [3] and, in a higher-order setting, Di Gianantonio, Honsell and Lenisa [4] develop the programme; bigraphs [5] are its best-known application. This is the strand that supplies our label discipline, and [1] is the form in which we use it.

Remark 1 (Which bisimilarity). The equivalence Caires locates is induced on an action-labelled transition system for a particular calculus. The equivalence this note works with is the context-labelled one of §3. They are not the same relation, and we do not borrow the earlier statement as though it were ours: claim (B) is re-established for our relation in §7. Conflating them would be exactly the sort of suppressed index this note exists to expose.

3 Preliminaries

We recall what is used and nothing more. For graph-structured lambda theories see [20]; for the label discipline, [1], which is in preparation and on which this note is frankly downstream.

Definition 2 (GSLT). A *graph-structured lambda theory* is a triple $\mathbb{G} = (\Sigma, E, R)$ where Σ is the signature of a lambda theory — so it may be many-sorted, constructors may bind, and terms may contain variables — E is a set of equations, and R is a set of rewrite rules, the graph of rewrites being adjoined to the structure rather than the structure being enriched over graphs. We write $\mathcal{T}(\Sigma)$ for the terms and $=_E$ for the least congruence containing E .

Definition 3 (Interactive). $\mathbb{G}$ is *interactive* when Σ carries a distinguished binary constructor K , the *interaction constructor* or *cut*, such that the left-hand side of every rule in R is of the form $K(l_1, l_2)$ up to $=_E$.

Parallel composition plays this role for CCS, the π -calculus, the ambient calculus and the rho calculus; application plays it for the lambda calculus.

3.1 Labels are minimal enabling contexts

Following [2] in the open form of [1]: a transition $t \xrightarrow{c} d[r]$ is asserted when $l \rightarrow r$ is a rule and $d[l] = c[t]$ is an idempotent relative pushout — so that c is a *least* context enabling the rewrite, minimality being a universal property in the relevant slice category. Leifer and Milner’s reactive systems fix a distinguished object as the domain of the IPOs, effectively restricting to ground terms; [1] observes that the proofs do not depend on it and adopts an open version, which is what we need, since a lambda theory’s terms carry variables.

Minimality is not a technical convenience. It is what keeps a label from carrying the whole term. A transition system labelled by *arbitrary* enabling contexts would be so informative that matching labels would nearly force syntactic agreement, and the equivalence it induced would already be at or near $=_E$ — collapsing the distinction this note exists to draw.

Hypothesis 4 (Redex RPOs). $\mathbb{G}$ has all redex-RPOs, so that the derived transition system exists.

We inherit this from [1], where establishing it for the theories of interest is work in progress. Everything below is conditional on it.

3.2 Matching labels

In a higher-order setting a minimal enabling context still carries *processes* — for the rho calculus the labels are of the form $- \mid \text{in}(n, \lambda x.q)$ with q an arbitrary process. Matching such labels syntactically yields a relation known to be too fine [6]; the standard remedy is to compare the payload up to the bisimilarity being defined.

Definition 5 (Context-labelled bisimulation). A symmetric relation $\mathcal{R}$ is a *context-labelled bisimulation* when $P \mathcal{R} Q$ implies: for every transition $P \xrightarrow{c} P'$ there are Q' and a label c' with $Q \xrightarrow{c'} Q'$, $P' \mathcal{R} Q'$, and $c \mathcal{R}^* c'$, where $\mathcal{R}^*$ compares labels position by position, relating the constructor skeletons by identity and the process payloads by $\mathcal{R}$. Bisimilarity $\sim_{\mathbb{G}}$ is the greatest such relation.

Lemma 6 (Well-definedness). *The functional whose greatest fixed point is Definition 5 is monotone, so $\sim_{\mathbb{G}}$ exists.*

Proof. $\mathcal{R}$ occurs only positively: in the conclusion of the matching clause, and in $\mathcal{R}^*$, which is built from $\mathcal{R}$ by identity and componentwise conjunction. A monotone functional on the complete lattice of symmetric relations has a greatest fixed point by Knaster–Tarski. $\square$

Hypothesis 7 (Congruence). A presentation is *admissible* when it comes equipped with a class $\mathcal{A}$ of contexts, containing those built from the rewrite constructors, such that $\sim_{\mathbb{G}}$ is a congruence with respect to $\mathcal{A}$, and observations are restricted to $\mathcal{A}$. We consider only admissible presentations.

This is a hypothesis, not a theorem, and deliberately so. We are interested only in context-labelled transition graphs that make bisimulation a congruence; barbed bisimulation, which obtains congruence by quantifying over contexts and a calculus-specific notion of observable, does not generalise across interactive GSLTs and we do not use it. Deriving labels from contexts so that congruence holds by construction is precisely the point of [2].

Remark 8 (The restriction is real). $\mathcal{A}$ is not all contexts. For the rho calculus, [1] proves congruence under $\text{out}(n, -)$, $\text{in}(n, -)$, $- \mid -$ and $\text{@}(-)$, and must exclude contexts with a hole *under* the quote, such as $\text{out}(\text{@}(-), z)$: communication turns on name equality, not on bisimilarity, so $p \sim q$ with $p \neq q$ gives $\text{@}p \neq \text{@}q$ and the two contexts behave differently. §10.1 returns to what this costs the present construction.

3.3 The generated logic

Definition 9 (The generated logic). $\mathcal{L}(\mathbb{G})$ has formulae

$$\varphi ::= \top \mid \neg\varphi \mid \varphi \wedge \psi \mid C(\varphi_1, \dots, \varphi_n) \mid \langle c \rangle \varphi$$

with one *structural* connective per n -ary constructor $C \in \Sigma$ and one *behavioural* modality per label. Satisfaction for the structural connectives is

$$P \models C(\varphi_1, \dots, \varphi_n) \iff P =_E C(t_1, \dots, t_n) \text{ with } t_i \models \varphi_i \text{ for each } i,$$

and $P \models \langle c \rangle \varphi$ iff $P \xrightarrow{c} P'$ with $P' \models \varphi$. We write $\mathcal{L}_s(\mathbb{G})$ for the structural fragment and $=_{\mathcal{L}}$ for logical equivalence.

Because K is typically subject to associativity, commutativity and a unit in E , the connective $K(\varphi, \psi)$ is a separating conjunction: it holds of P exactly when P admits *some* decomposition into two parts satisfying φ and ψ . The existential over decompositions is the whole difficulty. It is not a step, so no modality expresses it, and that is where (B) comes from.

3.4 Admissible equational theories

The results below need E to admit a well-founded notion of decomposition. We isolate the conditions rather than discover them mid-proof.

Definition 10 (Measure). For $P \in \mathcal{T}(\Sigma)$ let $|P|$ be the number of constructor occurrences and $\mu(P) = \min\{|Q| : Q =_E P\}$. A representative attaining the minimum is *lean*.

Definition 11 (Admissible E). E is *admissible* when:

- (E1) **Well-founded.** Every $=_E$ -class of closed terms contains a lean representative.
- (E2) **Hereditary.** If $C(\vec{t})$ is lean then each t_i is lean, and for $n \geq 1$ we have $\mu(t_i) < \mu(C(\vec{t}))$.
- (E3) **Sorted.** E relates only terms of the same sort.

Remark 12 (Why (E2) is not free). An earlier draft asserted that E is size-respecting on the presentations of interest. It is not: the unit law $P \mid \mathbf{0} = P$ is not size-respecting, and it is one of the equations doing the most work below, since it is what makes $P \mid \mathbf{0}$ one of the decompositions the separating conjunction quantifies over. (E1)–(E2) are the repair. They say nothing about how many decompositions a term has — possibly infinitely many — only that one may always descend along a lean one. Associativity, commutativity and unit for K satisfy them.

4 The observer extension

Fix an admissible interactive GSLT $\mathbb{G} = (\Sigma, E, R)$ with cut K , satisfying Hypotheses 4 and 7, with E admissible in the sense of Definition 11.

4.1 Argument formers, freely adjoined

K is binary; constructors have arbitrary arity and heterogeneous argument sorts. To present the arguments of an n -ary constructor to K they must be gathered into one operand.

Remark 13 (Not lists, and not Turing completeness). An earlier draft encoded the arguments as a list and appealed to Turing completeness to supply the list former. That was wrong twice over. Computability guarantees the ability to represent data; it does not supply an internal coding function injective modulo an arbitrary E , and injectivity modulo $=_E$ is exactly what the reconstruction below needs. And a general lambda theory is many-sorted, so a homogeneous list cannot hold heterogeneous arguments without a universal code sort, which is a further construction choice.

The repair is simpler than the thing repaired: *freely adjoin* an argument former per signature profile. Free adjunction gives correct sorting and injectivity by construction, introduces no encoding choice, and removes Turing completeness from the note entirely. Nothing below uses it.

Definition 14 (Observer extension). Let $\mathcal{O} \subseteq \Sigma$ be an *observation signature*. The *observer extension* $\text{Obs}(\mathbb{G}, \mathcal{O}) = (\Sigma^{\mathcal{O}}, E, R^{\mathcal{O}})$ is given by:

- for each $C \in \mathcal{O}$ of profile $\tau_1 \times \cdots \times \tau_n \rightarrow \tau$, a fresh sort A_C and a freely adjoined former $\text{args}_C : \tau_1 \times \cdots \times \tau_n \rightarrow A_C$;

- nullary *probe atoms* Ask_C , Build_C and $\text{Get}_{C,i}$ for $1 \leq i \leq n$;
- K overloaded so that $K(\text{Ask}_C, t)$, $K(\text{Get}_{C,i}, a)$ and $K(\text{Build}_C, a)$ are well formed for $t : \tau$ and $a : A_C$;
- E unchanged;
- $R^\mathcal{O} = R \cup \{ \text{ask}_C, \text{get}_{C,i}, \text{bld}_C : C \in \mathcal{O} \}$ where

$$\begin{aligned} \text{ask}_C : K(\text{Ask}_C, C(x_1, \dots, x_n)) &\longrightarrow \text{args}_C(x_1, \dots, x_n), \\ \text{get}_{C,i} : K(\text{Get}_{C,i}, \text{args}_C(x_1, \dots, x_n)) &\longrightarrow x_i, \\ \text{bld}_C : K(\text{Build}_C, \text{args}_C(x_1, \dots, x_n)) &\longrightarrow C(x_1, \dots, x_n). \end{aligned}$$

Rules of R are *proper*; the rest are *administrative*. We write $\mathbb{G}^+$ for $\text{Obs}(\mathbb{G}, \Sigma)$.

Four comments on the shape of this definition.

First, the overloading of K rather than the introduction of a fresh eliminator is the point of the whole construction. K is not an arbitrary constructor; it is the one the transition relation is indexed on, the one that defines what it means for two things to meet. A probe placed in interaction position turns a structural question into a rendezvous, and a rendezvous is what a derived transition system is built to observe.

Second, every administrative rule is *interaction-headed by construction*: its left-hand side is literally a K . An earlier draft put the constructor being opened on the left of the arrow, which forced an appeal to a unit for K and a special case, patched by an extra wrapper, for cuts without one — and the patch destroyed the very lemma the proof depended on. Putting the probe on the left removes the case distinction: the observer supplies the context, so no unit is needed anywhere.

Third, the probe atoms are indexed. It is Ask_C rather than a single Ask that makes the label of an opening transition record *which* constructor was found, and $\text{Get}_{C,i}$ rather than a single Get that makes projection record *which* argument was taken. This is what carries the information the reconstruction needs; see Remark 19.

Fourth, bld_C is not used in any proof below. It is present so that the extension supplies construction as well as deconstruction, and so that the observer's instruments form a coherent kit rather than a one-way mirror.

Proposition 15 ($\text{Obs}(\mathbb{G}, \mathcal{O})$ is an interactive GSLT). *$\text{Obs}(\mathbb{G}, \mathcal{O})$ is a GSLT, is interactive with the same cut K , and reflects proper transitions: for $P, P' \in \mathcal{T}(\Sigma)$ and $\rho \in R$, $P \xrightarrow{c} P'$ in $\mathbb{G}$ iff the same holds in $\text{Obs}(\mathbb{G}, \mathcal{O})$.*

Proof. Every rule of $R^\mathcal{O}$ has a left-hand side of the form $K(-, -)$: those of R by hypothesis, the administrative ones by construction. Reflection of proper transitions holds because $R^\mathcal{O} \supseteq R$ and the adjoined formers are mentioned in no rule of R , so no R -redex is created or destroyed by their presence, and the RPOs computed for R -redexes are unchanged. $\square$

Remark 16 (Observation as a separate label class). Nothing forces the administrative rules to be treated as ordinary rules. One may instead take them to constitute a distinct class of labels, matched separately. Because Definition 14 makes them interaction-headed anyway, the two treatments agree here, and we take the simpler one. The distinction would matter for a version of the construction in which opening did not require a probe.

Remark 17 (Equations as rewrite pairs). One may prefer a presentation with no equations, orienting each equation of E in both directions and adding both to the administrative rules. The two presentations agree up to administrative steps. We keep E as equations because doing so localises the source of nondeterminism: with E retained, the branching of ask_C comes from E -matching, which is exactly the existential over decompositions that the separating conjunction quantifies. It is worth stressing that these must be *pairs*. Orienting the equations in one direction only would let bisimilarity fall below $=_E$ onto raw syntax, which overshoots: the structural connectives respect $=_E$, so a relation finer than $=_E$ is finer than the logic and adequacy fails from the other side.

5 Reconstruction

5.1 Exposure

Lemma 18 (Exposure). *Let $P \in \mathcal{T}(\Sigma^\mathcal{O})$ and $C \in \mathcal{O}$ be n -ary. Then*

$$P \xrightarrow{\text{K}(\text{Ask}_C, -)} \text{args}_C(t_1, \dots, t_n) \quad \text{iff} \quad P =_E C(t_1, \dots, t_n),$$

and $\text{K}(\text{Ask}_C, -)$ is a least context enabling ask_C at P . Consequently the Ask_C -labelled transitions of P enumerate exactly the top-level E -decompositions of P at C . Likewise $\text{args}_C(\vec{t})$ has the single transition $\xrightarrow{\text{K}(\text{Get}_{C,i}, -)} t_i$ with that label.

Proof. The redex $\text{K}(\text{Ask}_C, C(\vec{x}))$ is formed from P by the context $\text{K}(\text{Ask}_C, -)$ exactly when $P =_E C(\vec{t})$; no smaller context enables it, since the rule's left-hand side requires the probe, and no larger one is minimal. Matching is up to $=_E$, which is where the enumeration comes from. For get, args_C is freely adjoined, so $\text{args}_C(\vec{t}) =_E \text{args}_C(\vec{s})$ iff $t_i =_E s_i$ for all i , and the projection is deterministic. $\square$

This is the lemma that does the work, and it is worth pausing on what it says in the case that motivated the difficulty. Take $C = \text{K}$ with E containing associativity, commutativity and unit. Then P 's Ask_K -labelled transitions are in bijection with the ways of writing P as $Q \mid R$ — including $P \mid \mathbf{0}$ and $\mathbf{0} \mid P$, and every rebracketing and permutation of a parallel product. The existential over splits in $P \models \varphi \mid \psi$ has become an existential over transitions. Everything else is bookkeeping.

Remark 19 (The label, not the target). An earlier draft said that the atom appearing “in the label” records the constructor. It did not: the atom was in the target, and only the rule name distinguished the constructors. With the probe on the left of the cut the claim is now literally true — Ask_C occurs in the context $\text{K}(\text{Ask}_C, -)$, which is the label — and the proof below may use it directly.

Lemma 20 (Calibration). *If $P =_E Q$ then $P \sim_{\text{Obs}(\mathbb{G}, \mathcal{O})} Q$.*

Proof. Transitions are defined up to $=_E$, so $=_E$ -equal terms have identical labelled transition sets, and the identity relation on $=_E$ -classes is a bisimulation. $\square$

Trivial as stated, and stated anyway, because it fixes the *lower* bound of the calibration: the enlargement must not push bisimilarity below the equational theory, or it overshoots the logic. It is also the property the alternative presentation of Remark 17 must be checked against.

5.2 The theorem

Theorem 21 (Reconstruction). *Let $\mathbb{G}$ be admissible with E admissible, and let $\mathcal{O} = \Sigma$. Then for closed $P, Q \in \mathcal{T}(\Sigma)$,*

$$P \sim_{\mathbb{G}+} Q \iff P =_E Q.$$

Proof. ($\Leftarrow$) is Lemma 20.

($\Rightarrow$) By strong induction on $\mu(P)$, with induction hypothesis: for all closed P', Q' with $\mu(P') < \mu(P)$, if $P' \sim_{\mathbb{G}+} Q'$ then $P' =_E Q'$. Note that the hypothesis constrains only the left argument, which is what allows the right-hand witnesses produced below to be arbitrary representatives rather than lean ones.

Let $P \sim_{\mathbb{G}+} Q$ and choose a lean representative $C(t_1, \dots, t_n)$ of P , which exists by (E1). By Lemma 18,

$$P \xrightarrow{K(\text{Ask}_C, -)} \text{args}_C(t_1, \dots, t_n).$$

Since $P \sim_{\mathbb{G}+} Q$, the term Q must match this step with a label related to $K(\text{Ask}_C, -)$. That label has no process payload — Ask_C is a nullary atom and the hole is the only other position — so label matching is identity on the constructor skeleton, and Q 's matching label is $K(\text{Ask}_C, -)$ itself. Distinct constructors carry distinct atoms, so no other opening rule could have matched. By Lemma 18 again, $Q =_E C(s_1, \dots, s_n)$ for some $\vec{s}$, and

$$\text{args}_C(t_1, \dots, t_n) \sim_{\mathbb{G}+} \text{args}_C(s_1, \dots, s_n).$$

Now apply $\text{Get}_{C,i}$. By Lemma 18 each side has exactly one transition with label $K(\text{Get}_{C,i}, -)$, to t_i and s_i respectively, so $t_i \sim_{\mathbb{G}+} s_i$ for each i .

By (E2) each t_i is lean and, for $n \geq 1$, $\mu(t_i) < \mu(P)$. The induction hypothesis gives $t_i =_E s_i$, whence $P =_E C(\vec{t}) =_E C(\vec{s}) =_E Q$. For $n = 0$ the constructor is nullary and $P =_E C =_E Q$ directly. $\square$

Remark 22 (Domains). Theorem 21 is a statement about $\sim_{\mathbb{G}+}$ restricted to closed Σ -terms. The relation itself is defined on all of $\mathcal{T}(\Sigma^\Sigma)$, where it also identifies, for instance, $\text{args}_C(\vec{t})$ with $\text{args}_C(\vec{s})$ exactly when the components agree. We are careful about this in §5.4, where an earlier draft was not.

Remark 23 (Infinite branching). A term with replication has infinitely many E -decompositions and therefore infinitely many Ask_K -labelled transitions. Nothing in the proof requires image-finiteness: the induction is on μ , not on transition depth, and no Hennessy–Milner style characterisation is invoked. §6 is what makes this safe on the logical side as well.

5.3 Binding

Remark 24 (Scope). For a binding constructor the arguments are abstractions rather than closed terms, and Theorem 21 as proved does not cover them: it would need α -equivalence to be part of E and (E1)–(E2) to hold of the resulting classes, a notion of transition beneath a binder, and a statement of what $K(\text{Ask}_C, -)$ means when C binds. We expect the lambda-theoretic setting to supply all three — an abstraction is a term of the theory, so the induction has somewhere to go — but expecting is not proving, and an earlier draft asserted this in one sentence. We therefore state the theorem for the first-order fragment and record the binding case as Problem 1. This is the largest gap between what the note proves and what it would like to prove.

5.4 Idempotence

Proposition 25 (Idempotence). $\sim_{\mathbb{G}^{++}}$ and $\sim_{\mathbb{G}^+}$ agree on $\mathcal{T}(\Sigma^\Sigma)$.

Proof. Since $\mathbb{G}^{++}$ has more rules than $\mathbb{G}^+$, monotonicity (Proposition 32) gives $\sim_{\mathbb{G}^{++}} \subseteq \sim_{\mathbb{G}^+}$. For the converse we exhibit a bisimulation rather than invoke Theorem 21 on both sides, which would be a domain mismatch: the theorem speaks of Σ -terms under $\mathbb{G}^+$, not of Σ^Σ -terms under $\mathbb{G}^{++}$.

Take $\mathcal{R} = \sim_{\mathbb{G}^+}$ and check the additional transitions. The constructors adjoined by Definition 14 are of two kinds. The probe atoms are nullary: their ask rules have targets $\mathbf{args}_{\text{Ask}_C}()$ with no components, so the transition carries no information beyond the identity of the atom, which $\sim_{\mathbb{G}^+}$ already fixes, since $\mathbf{K}(\text{Ask}_C, -)$ appears in labels. The argument formers $\mathbf{args}_C$ are not nullary, but the $\mathbf{Get}_{C,i}$ rules of $\mathbb{G}^+$ already project every one of their components, so the second-round $\text{ask}_{\mathbf{args}_C}$ transition is matched whenever the first-round projections are, and adds no separating power. Hence $\sim_{\mathbb{G}^+}$ satisfies the $\mathbb{G}^{++}$ matching conditions and is contained in $\sim_{\mathbb{G}^{++}}$. $\square$

This is not a side remark. It is the reason *Obs* is a construction rather than the first stage of a colimit, and it turns on a design decision that could have gone the other way: the probes are nullary and the argument formers are exhaustively projected. Introduce a family of n -ary destructors with partial projections instead and the ladder is real.

6 Separation

The operational side is done. The logical side is a separate question, and an earlier draft imported it from the literature at a generality the cited results do not support: those are theorems about particular spatial logics and particular calculi, and since the point of this note is to generalise past the calculus-by-calculus treatment, borrowing the key step defeats the purpose. We prove what is needed directly.

Theorem 26 (Structural separation). *Let E be admissible. Then for closed $P, Q \in \mathcal{T}(\Sigma)$,*

$$P =_{\mathcal{L}_s(\mathbb{G})} Q \iff P =_E Q,$$

where $\mathcal{L}_s(\mathbb{G})$ is the structural fragment.

Proof. $(\Leftarrow)$ Satisfaction of structural connectives is defined up to $=_E$.

$(\Rightarrow)$ We construct a characteristic formula. For closed P , by (E1) choose a lean representative $C(t_1, \dots, t_n)$ and set

$$\chi_P := C(\chi_{t_1}, \dots, \chi_{t_n}),$$

which is well defined by induction on μ , using (E2) for the descent, with $\chi_P := C$ for nullary C .

We claim $Q \models \chi_P$ iff $Q =_E P$. If $Q \models C(\chi_{t_1}, \dots, \chi_{t_n})$ then $Q =_E C(s_1, \dots, s_n)$ with $s_i \models \chi_{t_i}$, so by induction $s_i =_E t_i$, so $Q =_E C(\vec{t}) =_E P$. Conversely $P \models \chi_P$ by construction. Hence if $P \neq_E Q$ then χ_P separates them. $\square$

Remark 27 (What this buys). Three things. The note becomes self-contained: no separation result is imported. The image-finiteness objection dissolves, because the argument goes from the logic to $=_E$ directly by characteristic formulae and never passes through a Hennessy–Milner characterisation of bisimulation, which is where image-finiteness or infinitary conjunction would

have been needed. And the hypotheses are visible: separation holds exactly for admissible E , so a presentation whose equational theory fails (E1)–(E2) is one for which the note claims nothing. Note also that no boolean structure was used — χ_P is purely structural.

Corollary 28 (Logical correspondence). *Under the hypotheses of Theorems 21 and 26, for closed Σ -terms*

$$=_{\mathcal{L}_s(\mathbb{G})} = =_E = \sim_{\mathbb{G}^+} .$$

The structural fragment generated from $\mathbb{G}$ is adequate for context-labelled bisimulation in $\mathbb{G}^+$.

7 Non-vacuity

Corollary 28 would be worthless if $\sim_{\mathbb{G}}$ were already $=_E$, since the construction would then close a gap that was not there. Remark 1 also owes the reader claim (B) for *our* relation rather than the literature’s. Both debts are settled by the same observation.

Proposition 29 (The gap is generically nonempty). *Let $\mathbb{G}$ contain two distinct nullary constructors c, d of the same sort occurring in no rule of R and in no equation of E . Then $c \sim_{\mathbb{G}} d$ and $c \not\equiv_E d$, so $\sim_{\mathbb{G}^+} \subsetneq \sim_{\mathbb{G}}$.*

Proof. Neither c nor d occurs in any left-hand side, so neither contributes a redex, and for any context D the transitions of $D[c]$ and $D[d]$ are in label-preserving bijection with those of $D[-]$ ’s own redexes. Hence $\{(c, d)\} \cup =_E$ extends to a bisimulation. They are $=_E$ -distinct by hypothesis, so by Theorem 21 they are not $\sim_{\mathbb{G}^+}$ -related, witnessed by $K(\text{Ask}_c, -)$. $\square$

The example is deliberately austere: it shows that the gap is nonempty for essentially any signature carrying inert data, without relying on a coincidence of the equational theory. Richer witnesses are presentation-dependent, and it is worth being explicit that they are.

Example 30 (Replication, and a caveat). In a presentation with replication as a primitive constructor and E *not* containing the unfolding law, $!P$ and $!P \mid P$ are behaviourally indistinguishable and $=_E$ -distinct, so they witness the gap. If E does contain the unfolding law they are $=_E$ -equal and witness nothing. The size of the gap is a property of the presentation, not of the calculus informally described.

Remark 31 (Why minimality matters here). Proposition 29 depends on labels being minimal. Were labels arbitrary enabling contexts, a label would carry essentially the whole surrounding term, matching would force near-syntactic agreement, and $\sim_{\mathbb{G}}$ would sit at or near $=_E$ — leaving no gap to close and no content to Corollary 28. The choice of label discipline is therefore not a technical preliminary but a load-bearing part of the statement.

8 The dial

Nothing forces the construction to be run on the whole signature, and the point of Definition 14 taking $\mathcal{O}$ as a parameter is that the observer’s instruments are part of the data.

Proposition 32 (Monotonicity). *If $\mathcal{O} \subseteq \mathcal{O}'$ then $\sim_{\text{Obs}(\mathbb{G}, \mathcal{O}')} \subseteq \sim_{\text{Obs}(\mathbb{G}, \mathcal{O})}$, with $\sim_{\text{Obs}(\mathbb{G}, \emptyset)} = \sim_{\mathbb{G}}$ and, under the hypotheses of Theorem 21, $\sim_{\text{Obs}(\mathbb{G}, \Sigma)} = =_E$ on closed Σ -terms.*

Proof. The administrative rules of $\text{Obs}(\mathbb{G}, \mathcal{O}') \setminus \text{Obs}(\mathbb{G}, \mathcal{O})$ carry fresh probe atoms, hence fresh labels, and add transitions that must be matched; the largest relation satisfying more conditions is contained in the largest relation satisfying fewer. The endpoints are Proposition 15 and Theorem 21. $\square$

This is the sharp form of the resolution promised in §1. There is no single observational bisimulation against which a spatial logic must be measured. Observational equivalence is indexed by the destructuring capabilities admitted to the observer, and (A) and (B) name the two ends of the resulting family.

What does *not* follow, and what an earlier draft asserted as an immediate corollary, is that the fragment of the logic retaining structural connectives only for $\mathcal{O}$ is adequate for $\sim_{\text{Obs}(\mathbb{G}, \mathcal{O})}$.

Conjecture 33 (Partial characterisation). *For $\mathcal{O} \subseteq \Sigma$, $\sim_{\text{Obs}(\mathbb{G}, \mathcal{O})}$ coincides with logical equivalence in the fragment of $\mathcal{L}(\mathbb{G})$ whose structural connectives are those of $\mathcal{O}$.*

The obstacle is real. A formula may inspect an opened constructor sitting underneath an unopened one; equations may relate terms whose roots lie on opposite sides of $\mathcal{O}$; and a binder may move material across an apparent structural boundary. None of that is handled by relativising Theorem 21. We prove one endpoint-adjacent case and leave the rest.

Proposition 34 (Downward closure). *Conjecture 33 holds when $\mathcal{O}$ is closed under the subterm relation of lean representatives — that is, when every constructor that can occur beneath a constructor of $\mathcal{O}$ in a lean term is itself in $\mathcal{O}$.*

Proof. Under that condition the descent in the proofs of Theorems 21 and 26 never leaves $\mathcal{O}$, so both arguments run verbatim with Σ replaced by $\mathcal{O}$ and $=_E$ replaced by its restriction to the $\mathcal{O}$ -generated subalgebra. $\square$

We suspect Conjecture 33 is the most interesting statement in the vicinity, and that its proof would say something about how partial structural knowledge composes. We do not give it away as a corollary.

9 Two cautions for downstream use

9.1 Transitions and derivations

Lemma 18 says the Ask_C -labelled transitions of P enumerate its decompositions. The transition relation is extensional: $P \xrightarrow{c} P'$ holds when a witnessing E -match exists. If two distinct E -matches yield the same label and the same target up to $=_E$, they are one transition and two derivations.

For bisimulation this is immaterial — only existence is used. For any quantitative lift it is not: weights and costs attach to derivations, and a construction that collapses derivational multiplicity will misprice. Anyone composing this note with a weighted or cost-accounted refinement should carry the derivation structure explicitly rather than the relation.

9.2 Cost is bounded by depth, not equal to it

In a cost-accounted GSLT [23] every rewrite carries a price, and administrative rules are rewrites like any other. It is tempting to conclude that the price of evaluating a structural formula of

depth d is a function of d , so that pricing schedules stipulated elsewhere become derived step counts. That is too quick. Nesting depth bounds the length of the longest *sequential* destructuring path, but the cost of checking depends further on branching: a conjunction requires two subchecks, a negation may require exhausting a search, and a separating conjunction may enumerate many E -decompositions before finding one that works, or all of them before concluding that none does. Sharing and caching move the figure again.

The defensible statement is that depth lower-bounds and structurally bounds sequential destructuring depth, while actual checking cost depends additionally on branching and on the evaluation strategy for the boolean layer. That is more useful downstream than an identity that would not survive contact with an implementation.

10 The construction is not capability-safe

We state this plainly rather than hedging it, because a reader who knows object-capability discipline will see it in Definition 14 and will want to know whether we noticed.

Unrestricted opening destroys encapsulation. Any observer may form $K(\text{Ask}_C, -)$, so any term whatever may be taken apart, and $\text{Get}_{C,i}$ then yields the arguments as data. In a calculus where an opaque reference is the unit of authority, this is fatal: open the reference and harvest whatever authority its contents carry. Confinement fails, and with it every argument that depends on it.

The repair is known in outline. Mike Stay’s suggestion is that constructors and destructors should each take a capability token, so that the right to build and the right to open are held rather than ambient. There are two readings — a *permission*, licensing the rule, or a *brand*, so that building stamps the term and opening checks the stamp — and the second is the more useful, since it makes legibility a property of the term rather than a global property of the world. The token must gate the whole kit rather than opening alone, since an observer able to take an administrative step should be able to take its converse.

Tokens require a factory, and two observations are worth recording before we stop.

Remark 35 (The factory is an import). Remark 13 obtained the argument formers by free adjunction. That cannot be done again here. Unforgeability is not a claim about what can be computed, nor about what can be adjoined, but about what a context can *write down*: in a signature of freely adjoined constructors every term is writable by anybody, which is precisely why free adjunction was safe for args_C and is useless for a token. A genuine source of privacy must be imported — restriction, in the general case, giving a factory of the shape $\text{let } c = \text{freshCap}() \text{ in } P$, essentially $(\nu c)P$. Reflective calculi, in which a name is a quoted process, may be able to discharge the import internally; whether quotation *originates* privacy or merely propagates it from a seed we do not settle.

Remark 36 (Factories resist the construction). Whatever the factory is, Definition 14 must not be applied to it. Give it an opening rule and a token can be taken apart; if it can be taken apart it can be rebuilt; if it can be rebuilt it can be forged; and if it can be forged nothing is a capability. So the capability-safe version of the construction is not “the observer extension, with a caveat” but “the observer extension on $\mathcal{O} \subseteq \Sigma \setminus F$, where F is the minting apparatus” — and the exclusion is what makes the rest of it safe. An earlier draft said the factory is the *unique* constructor admitting no destructor. That is more than we can support: a system may carry several fresh-name generators, sealed abstract types, or independent minting authorities, and in

nominal presentations freshness is a binder rather than a constructor at all. The property that matters is not uniqueness but that the mechanism generating unforgeable authority is not itself generically destructible by the observer extension.

10.1 Reflection, and an inherited restriction

Remark 8 recorded that congruence in the ρ calculus holds only for a restricted class of contexts, excluding holes beneath the quote, because communication turns on name equality rather than on bisimilarity. That restriction is inherited here and constrains $\mathcal{O}$.

The opening rule for the quote, $K(\text{Ask}_@, @(p)) \rightarrow \text{args}_@(p)$, does not itself place a hole beneath $@$: the hole is beside the quote, not inside it. So we do not expect it to disturb the congruence result. But the observation is a genuine addition to the context class, and whether $\mathcal{A}$ extended by the administrative contexts still supports congruence is a question for [1]’s machinery rather than for assertion here. We flag it as Problem 3 and note that a reader wanting a safe default may take $\mathcal{O}$ to exclude the quote, at the cost of leaving the name predicate outside the correspondence of Corollary 28.

11 What this settles, and for whom

Adequacy claims survive contact with intensionality results. A generated spatial-behavioural logic may be asserted adequate, provided the assertion names the relation: bisimulation in the observer extension for a stated $\mathcal{O}$, not in the base. Assertions that omit the index are not wrong so much as ambiguous, and the ambiguity is what invites the objection.

Structural observation is representable as interaction. Not identical to it. The distinction matters: the enlargement adds transitions whose purpose is to expose structure, and an architecture that appeals to this construction should say that its observer holds instruments, not that its instruments were never instruments.

Three restrictions, kept apart. An architecture in which an agent observes its environment faces three different limits, and collapsing them causes trouble. There is the *ideal observer* with the full observation signature, whose reach is $=_E$. There is the *capability-limited* observer, holding some $\mathcal{O} \subsetneq \Sigma$, whose reach is the corresponding point of Proposition 32. And there is the *budget-limited* observer, who holds instruments it cannot afford to use, whose reach is smaller again and is not characterised here at all. Adequacy is a claim about the first.

Spatial and behavioural addresses coincide. In $\text{Obs}(\mathbb{G}, \mathcal{O})$ a decomposition site is a redex position, so constructions identifying a structural locus with a modal label are well typed rather than analogical. This does not supply an enabling theorem: a locus that is a redex position for an administrative rule need not correspond to a *proper* step reachable in $\mathbb{G}$, and claims about reachability still require their own argument.

Morphisms. Where the morphisms of a category of GSLTs preserve bisimulation [20], an assignment depending on spatial structure is not functorial, and the usual repair is to narrow the morphism class by hand to maps preserving the generated logic. Under Theorem 21 the

narrowing is not a narrowing: logic-preserving maps of $\mathbb{G}$ are exactly bisimulation-preserving maps of $\mathbb{G}^+$, and the repair becomes functoriality on the image of $\text{Obs}(-, \Sigma)$.

12 Open problems

1. **Binding.** Extend Theorem 21 to binding constructors: α -equivalence within E and the resulting measure, transitions beneath binders, and the meaning of a probe applied to an abstraction. See Remark 24.
2. **Conjecture 33.** The partial characterisation, beyond the downward-closed case of Proposition 34.
3. **Reflection and congruence.** Whether the admissible context class of [1] remains a congruence class when extended by the administrative contexts, in particular for the quote. See §10.1.
4. **Canonicity as a universal property.** We have called the construction uniform and presentation-directed, not canonical: it is mechanical in the signature, but $\mathcal{O}$ is a choice, the probe indexing is a choice, and the treatment of binders will be a choice. Is $\text{Obs}(-, \Sigma)$ a left adjoint — free among extensions making $\mathcal{L}_s(\mathbb{G})$ behavioural? An adjunction would upgrade “evident” to “forced”, which matters wherever the absence of a design choice is doing argumentative work.
5. **The capability-indexed family.** With tokens, bisimilarity becomes indexed by the capabilities an observer *holds*, an object of the calculus rather than a parameter of the metatheory. This is a sharper form of §8 and needs the factory of Remark 35. It also wants a no-amplification theorem: one cannot open one’s way to a token one was not given. That statement has a circular smell — unforgeability rests on the gate, the gate rests on tokens — and needs a well-foundedness argument.
6. **Weak equivalences.** Everything here is strong. Administrative steps are natural candidates for being made silent, and [1] records that weak bisimilarity is a congruence only for reactive contexts, so the weak theory needs its own treatment.
7. **Relation to the semantic route.** [17] obtains extensionality from partial failures. Is there a translation carrying that model into an observer extension, so that failures and administrative openings turn out to be two presentations of one mechanism?
8. **Precedent.** Reifying constructors as tags and supplying observation actions is elementary enough that we expect precedents in the literature on definability by experiments, applicative and environmental bisimulation, and observational equivalence for abstract data types. A targeted pass is owed before any priority claim is made.

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